



HfO₂ FeRAM Production System

Master Report

University of Pristina × Infineon Collaboration

Ontologies Development Workstream

Project GitHub: <https://github.com/OPXHS/Ontologies-Development>

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Abstract

Process Ownership

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Chapter 3

ALD HfO₂ Deposition

3.1 Module Purpose

The ALD HfO₂ deposition module converts a deposition-ready capacitor region into an electrode-bounded ferroelectric hafnia film whose composition, interfaces, and thermal history are explicit. Within the production-system boundary used here, the module receives the prepared wafer state from the CMOS front end, forms or accepts the bottom-electrode boundary, deposits the HfO₂-based layer by an ordered ALD cycle, completes the upper boundary, and applies any route-scoped crystallization treatment. The bottom electrode must precede HZO deposition, but the anneal can occur before or after the upper electrode according to the selected route [1–4].

Module success is a crystallized film with an orthorhombic ferroelectric contribution inside a functioning capacitor stack, not film deposition alone. ALD provides high-aspect-ratio coverage, yet the deposited material can remain polycrystalline and contain nonferroelectric phases, so geometry and nominal thickness do not establish the required phase state [5]. Electrode oxygen exchange, stress, anneal placement, and anneal temperature then determine whether the deposited layer reaches that state or develops monoclinic material, leakage paths, or shorts [1, 3, 6, 7]. The module hands the resulting capacitor-level material state to downstream integration with its route metadata intact: film composition and thickness, electrode identities and positions, ALD chemistry and sequence, exposure history, and thermal treatment.

3.2 Input Condition

The module assumes a wafer with a defined capacitor region and a surface state ready for the first in-scope preparation or bottom-electrode operation. That input record must preserve substrate or underlying-film identity, prior exposure or vacuum-break history, and any pre-existing interface layer, because a vacuum break is associated with increased carbon and tetragonal-phase stabilization, while deliberately inserted interlayers alter crystallization and defect migration [6]. Cleaning, seed-layer insertion, and bottom-electrode formation are therefore module operations unless a route explicitly records them as already complete.

The ledger does not define one complete incoming wafer state. Its seven input-condition claims come from three sources and mainly specify the intended HZO film—10 or 25 nm, nominal 1:1 Hf:Zr, and one 50–200 μm capacitor-diameter range—rather than the underlying CMOS surface [1, 8]. A TiN/HZO/TiN route records a 10 nm Hf_{0.5}Zr_{0.5}O₂ target [9]. The remaining eleven sources provide no claim under the input-condition target, so substrate material and incoming topography cannot be completed by inference.

3.3 The Shape of the Evidence

The literature reports routes, not recipes. A reported temperature, duration, pressure, or cycle count becomes interpretable only with the configuration that produced it: precursor and oxidant chemistry, reactor mode, film composition and thickness, electrode identities and positions, exposure history, anneal placement, and measurement protocol. The chapter therefore reports two evidence objects. A *literature window* is the conditioned span documented by several routes; a *route instance* is one value inside, at the edge of, or beyond that span. Neither object is a recommendation, and an instance never becomes a norm merely because it is precise.

The useful synthesis is directional coupling between variables. Electrode boundary condition determines anneal placement: an uncapped route anneals before the upper electrode, whereas a capped route anneals after it [5, 8]. Deposition temperature increases oxygen-vacancy indicators and dead-layer formation; the higher vacancy state then raises remanent polarization while bringing fatigue onset to a lower cycle count [10]. Precursor-purge duration changes retained carbon and nitrogen impurities and also shifts breakdown: reduced purge does not completely remove ligands, but extending the studied purge from 3 to 90 s changes the same capacitor series from no fatigue or breakdown through 10^9 cycles to breakdown at 6×10^7 cycles [11]. These observations establish directed dependencies without claiming that the intermediate impurity signal alone explains the breakdown direction.

Transfer between routes would require demonstrated equivalence of every variable that changes the relevant outcome. For a deposition value, that includes precursor identity and delivery mode, line and source temperatures, co-reactant exposure, reactor mode, substrate temperature, cycle architecture, composition, and thickness. For a thermal value, it additionally includes the electrode material, whether the film is capped, anneal position, ambient, ramp, duration, and preceding exposure. For an electrical result, stack area, conditioning state, waveform, field or voltage, frequency, and driven electrode must match. When those conditions are absent, the value remains route-scoped; when a condition is unreported, the gap remains an assertion rather than an invitation to infer it.

3.3.1 Invariant and Route-Variable Sequence

Bottom-electrode formation precedes HZO deposition in every route that states this relation. The invariant appears for sputtered TiN and W bottom electrodes and for wafer-scale SS/SL, Zr-rich, and W/HZO/W stacks [1–4]. No ledger claim reverses the relation. This supports a necessary **precedes** assertion in the later ontology rather than a preferred process order.

Anneal placement is route-variable because the electrode boundary at heating changes from one configuration to another. PDA is recorded as uncapped RTA and PMA as RTA after TiN capping; other routes place post-deposition annealing before the upper electrode or post-metallization annealing after it [5, 7, 10]. One study deliberately compares both orders, applying RTP before top-electrode deposition to its uncapped sample and after top-electrode deposition to its capped sample [8]. A wafer-scale route narrows placement further to after top-TiN patterning but before bottom-TiN localization [2]. These are route-scoped **precedes** relations; none can replace the bottom-electrode invariant.

3.3.2 First-Order and Second-Order Variables

First-order variables are those for which the ledger records a qualitative change of phase state or device survival. Anneal temperature and its placement relative to capping are first-order: top-electrode temperature and capped versus uncapped order change monoclinic or orthorhombic phase outcomes, while annealing W/HZO/W above 720 °C produces large leakage and shorts [1, 3, 8]. Deposition temperature is first-order because it changes whether an interfacial dead layer forms and shifts the vacancy-mediated polarization–fatigue state [10]. Precursor-purge

duration is first-order because the studied series crosses from survival beyond 10^9 cycles to dielectric breakdown, not merely between two numerical values [11]. Electrode material and oxygen affinity are first-order because oxygen-gettering TiN promotes the ferroelectric phase, while oxygen-supplying $\text{RuO}_2/\text{IrO}_2$ and oxidation-resistant alternatives change the reported endurance boundary [1, 6, 12]. Oxidant exposure is also first-order in the supported W/HZO/W route: short ozone pulses produce pinched loops, whereas a 15 s pulse produces an open loop [3].

Second-order variables shift a value inside a demonstrated working configuration without a ledger-supported qualitative transition. The precursor-source temperatures, ALD chamber pressure, sputter power, and exact cycle or supercycle count are second-order on the present evidence: the ledger records settings or numerical trends, but no isolated phase change or device-destruction threshold for those variables [3, 10, 11]. Electrode-deposition pressure is a position-specific exception rather than evidence that every pressure is first-order, because TiN top-electrode microtexture is reported to decide the resulting crystalline phase [6]. Gas-flow rates, anneal ambient independent of temperature and placement, and precursor delivery-mode effects remain unclassified because the ledger does not isolate their qualitative impact.

Evidence density and importance are independent. The first-purge target contains only three claims from one source, yet its purge-duration series produces one of the ledger’s largest outcome changes: endurance falls from beyond 10^9 to 6×10^7 cycles as the precursor purge extends from 3 to 90 s [11]. Sparse coverage therefore reduces confidence in transfer, not the observed effect within that route.

3.4 Pre-Deposition Preparation

3.4.1 Starting Film, Substrate, and Device Conditions

The starting-film specification constrains the cycle architecture that follows. A 10 nm, nominally equimolar HZO condition is paired with a 250°C wafer temperature and 50–200 μm capacitor diameters in one TiN route, whereas another route records a 25 nm HZO film without supplying the same device dimensions [1, 8]. A TiN/HZO/TiN route records a 10 nm $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ target [9]. The input-condition target contains seven claims from three of fourteen sources; it leaves substrate material, incoming topography, and prior thermal state unreported for the other eleven.

3.4.2 Surface Cleaning and Exposure Control

Surface cleaning establishes the chemical state on which the first in-scope interface or electrode is formed. Its process function is to remove residual contamination and to make exposure history explicit; the failure mode documented in the ledger is that a pre-deposition vacuum break increases carbon and stabilizes the tetragonal phase [6]. The operation therefore controls a boundary condition for crystallization rather than merely improving visual cleanliness.

The available cleans are purpose- and position-specific. One substrate route applies SPM followed by RCA SC-1/SC-2, while another uses sequential ultrasonic acetone, ethanol, and DI-water cleaning [7, 10]. Wet chemistry used after electrode formation serves a different operation and is not evidence for the incoming preclean [1, 11, 13]. Six of fourteen sources contribute surface-preclean claims; the remaining eight do not report the preclean chemistry or exposure control.

3.4.3 Interface and Seed-Layer Engineering

Interface layers direct crystallization and defect transport. ZrO_2 , Al_2O_3 , TiO_2 , and LSMO interlayers are reported below ferroelectric hafnia to alter crystallization, while Al_2O_3 barriers restrict defect migration or grain-boundary conduction [6, 12]. The same Al_2O_3 insertion

introduces voltage division and a depolarization field, so the barrier function and electrical penalty are coupled rather than independently selectable [6, 12].

Interface position determines which mechanism is available. Top, bottom, dual, and unseeded ZrO_2 configurations remain separate specimens; the dual layers are resolved at both HZO interfaces, and their phase and cycling response cannot be assigned to a generic “seeded” class [7]. Pt insertion suppresses the W/HZO interfacial layer and raises the interface barrier, whereas a Zr-rich middle layer divides HZO into a trilayer architecture [3, 4]. In Si-doped stacks, a thicker PEALD SiO_2 monolayer shifts the doping concentration relative to thermal ALD, so nominal cycle ratios do not assert equal composition [13]. Nine sources cover interface preparation; none supplies a cross-route equivalence test for these alternatives.

3.5 Electrode Integration

3.5.1 Bottom Electrode Formation

Bottom-electrode formation fixes the mechanical and oxygen-exchange boundary before HZO growth. Supported routes sputter TiN, W, or a TiN/Ru stack before oxide deposition; the SNAP route instead forms TiN and HZO sequentially in one ALD chamber without a vacuum break [2, 6, 8, 10]. The position is invariant even though material, method, and thickness vary.

A TiN/HZO/TiN route records a 15 nm bottom TiN layer sputtered at 100 W, 2.5 mTorr, with Ar and N_2 flows of 50 and 3 sccm [9]. The thickness lies inside the ledger’s 10–90 nm electrode envelope, the power lies between the reported 50 and 250 W instances, the pressure extends the sputter-pressure low end below 5.0 mTorr, and the gas-flow quantities have no prior sccm entry. These relations locate one route against the evidence window without transferring its settings.

3.5.2 Top Electrode and Capping

Top-electrode and capping formation determine the boundary condition present during subsequent heating. Some routes deposit a TiN, Ni, or W top electrode before crystallization, while the seed-layer PDA route anneals before its upper TiN is added [3, 7, 8, 10]. In the purge study, the W capping layer and the later Pt characterization contact are separate operations; Au/Pd hard masks and Ti/Au contacts likewise do not become capacitor electrodes merely because they are metallic [1, 2, 11].

A TiN/HZO/TiN route records a 15 nm sputtered TiN top electrode but no separate top-electrode power, gas flow, or pressure [9]. Its thickness is inside the 10–90 nm literature envelope, but the missing deposition conditions prevent the bottom-electrode settings from being copied to the top position.

3.5.3 Electrode Material Selection and Interface Consequences

Electrode oxygen affinity directs the film’s phase and defect boundary. TiN gettering and bottom-electrode oxidation promote the ferroelectric phase, while RuO_2 and IrO_2 supply oxygen and are associated with improved endurance relative to oxygen-gettering alternatives [1, 6]. W contributes a tensile-strain route through its low thermal-expansion behavior, and a W/Pt/HZO/W stack deliberately assigns top W and bottom Pt to combine mechanical stress with a sharper oxide interface [3, 6].

Electrode position also changes the mechanism for the same nominal material. TiN top-electrode microtexture depends on deposition pressure, and room-temperature TiN suppresses monoclinic formation through stress-induced crystallization, whereas deposition at 500 °C produces a confirmed monoclinic phase [1, 6]. Material identity, stack position, deposition condition, and anneal placement must therefore remain separate properties in the route record.

Table 3.1: Electrode integration routes and stack conditions

Stack position	Electrode	Integration method	Thickness	Process condition	Source/configuration
Three-dimensional capacitor	Not covered in sources	Conformal ALD	10 nm	High-aspect 1T–1C FeRAM	Song configuration [5]
Bottom and top	TiN	In-tool SNAP ALD	Not covered in sources	Without vacuum break	Roadmap configuration [6]
Bottom and top	TiN	DC sputtering	50 nm	Reference capacitor	Choi configuration [10]
Bottom	W	DC magnetron sputtering	Not covered in sources	3.3 W/cm ² ; 5.0 mTorr Ar	W baseline [11]
Top capping	W	DC magnetron sputtering	20 nm	Reduced deposition time	W capping configuration [11]
Top contact	Pt	Shadow-mask deposition	50 nm	Electrical characterization	Pt contact configuration [11]
Bottom and top	TiN	RF sputtering	90 nm	250 W; Ar:N ₂ 20:1; RT or 500°C	TiN temperature study [1]
Hard mask	Au/Pd	Sequential e-beam evaporation	85/3 nm	Sequential deposition	Hard-mask configuration [1]
Upper	TiN	RF sputtering	50 nm	400 μ m; 29 ohm/sq	Seed-layer device [7]
Bottom stack	TiN/Ru	Not covered in sources	20/100 nm	300 nm SiO ₂ base	Induced-crystallization capacitor [8]
Top	Ni	Sputtering	50 nm	50 W cluster system	Ni-top capacitor [8]
Bottom and top	TiN	Bottom ALD; top PVD	10 nm	Silicon-doped hafnia capacitor	Neuber configuration [13]
Bottom and top	TiN	Reactive sputtering	30 nm	Superlattice capacitor	Wafer-scale configuration [2]
Electrical contacts	Ti/Au	Evaporation and lift-off	Not covered in sources	Contacts to TiN electrodes	Wafer-scale configuration [2]
Bottom and top	W	Magnetron sputtering	30 nm	Room temperature; top 80×80 μ m ²	Zirconium-rich capacitor [4]
Bottom and top	W	Bottom sputter; top RF sputter	50 nm	Room temperature; 30×30–100×100 μ m ²	Wake-up-free capacitor [3]
Hybrid interfaces	Top W; bottom Pt	RTA-integrated stack	Not covered in sources	Mechanical stress with Pt interface	Hybrid capacitor [3]
Gate stack	Polysilicon/TiN	Not covered in sources	Not covered in sources	28 nm CMOS FeFET stack	FeFET configuration [14]

3.6 ALD Film Formation

3.6.1 Metal-Precursor Dose

Precursor identity determines the delivery conditions that can be interpreted together. The ledger spans cyclopentadienyl, alkylamide, TEMA-, and TDMA-based precursors; organometallic ligands are associated with residual impurities, and the alkylamide family carries a reported thermal-stability limit [5, 6]. In the most resolved TEMA-based PEALD route, bubbler temperatures,

hotter downstream manifolds, Ar-assisted dosing at 80 mTorr, and unequal Hf and Zr pulse lengths form one delivery system [11]. Separating any one number from that system removes the condensation-control and dose context.

One TiN/HZO/TiN route records TEMAH at 90 °C and TEMAZ at 110 °C for deposition at 250 °C [9]. The TEMAH temperature matches a prior 90 °C instance, but the TEMAZ temperature exceeds the previously documented 85 and 90 °C values [3, 11]. This is an extension, not a contradiction: the route omits delivery mode and line temperature, which are the conditions required to reconcile absolute source temperatures.

3.6.2 First Purge

The first purge is the cycle boundary that separates the metal-precursor dose from co-reactant admission. In the process model, its function is to clear the reaction space before the oxidant arrives; incomplete clearing is represented by retained precursor ligands, and the measured reduced-purge films retain carbon and nitrogen impurities [11]. This definition supplies the operation’s role without turning one experimental duration into a default.

Purge duration controls both retained chemistry and device survival in the only resolved series. Ar at 80 mTorr is held as the purge environment while duration spans 3–90 s; the 3 s device shows no fatigue or breakdown through 10^9 cycles, but the 90 s device breaks at 6×10^7 cycles [11]. Only one of fourteen sources reports the first purge as its own target, leaving thirteen without a first-purge duration or independent failure comparison.

3.6.3 Co-Reactant Dose

Co-reactant exposure determines how each adsorbed precursor half-cycle is oxidized. Thermal routes report H₂O or O₃, while plasma-enhanced routes report oxygen-plasma exposures; one PEALD sequence assigns different power–duration pairs to the Hf and Zr subcycles [1, 5, 7, 11]. In W/HZO/W, increasing the ozone pulse from below 5 s to 15 s changes the pristine response from pinched to open hysteresis and reduces carbon at the high-dose condition [3]. Oxidant identity and exposure must therefore remain attached to the metal subcycle and electrode stack.

The TiN/HZO/TiN route does not report its oxidant [9]. It is not evidence that the route used no oxidant, nor may H₂O be supplied by analogy. The recorded gap shows that an otherwise ordered route can leave a first-order cycle variable unspecified.

3.6.4 Second Purge

The second purge is the cycle boundary between the co-reactant exposure and the next metal-precursor dose. In the process model, it restores a separated starting state for repetition; failure to clear the preceding exposure would leave the next dose without the temporal isolation that defines the recorded ALD sequence. This is a functional operation definition, not a reported duration.

The evidence base identifies only nitrogen as the purge and carrier gas in one thermal HZO route [8]. One of fourteen sources therefore covers the second-purge target, while thirteen report no second-purge gas; all fourteen omit a second-purge duration, pressure, and measured failure mode. The chapter preserves that evidence gap rather than copying first-purge conditions into the second purge.

3.6.5 Cycle and Supercycle Architecture

Supercycle count is a dependent variable in the thermal HZO temperature series. Holding film thickness at 10 nm, the route uses 56, 61, and 68 supercycles as chamber temperature rises from 120 to 250 °C because growth per supercycle decreases with temperature [10]. The count encodes the deposition condition rather than specifying it independently.

Cycle architecture can change composition sequence while total count or thickness remains fixed. A nanolaminate series holds 120 combined $\text{HfO}_2/\text{ZrO}_2$ cycles while changing block length, the SS/SL comparison reaches 5.5 nm through alternating single cycles or grouped symmetric blocks, and discrete-feeding ALD repeats the precursor half-cycle before the reactant half-cycle [2, 6, 7]. A separate PEALD route uses a 3:2 Hf:Zr dose cycle to target nominal $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_{2-x}$ [11].

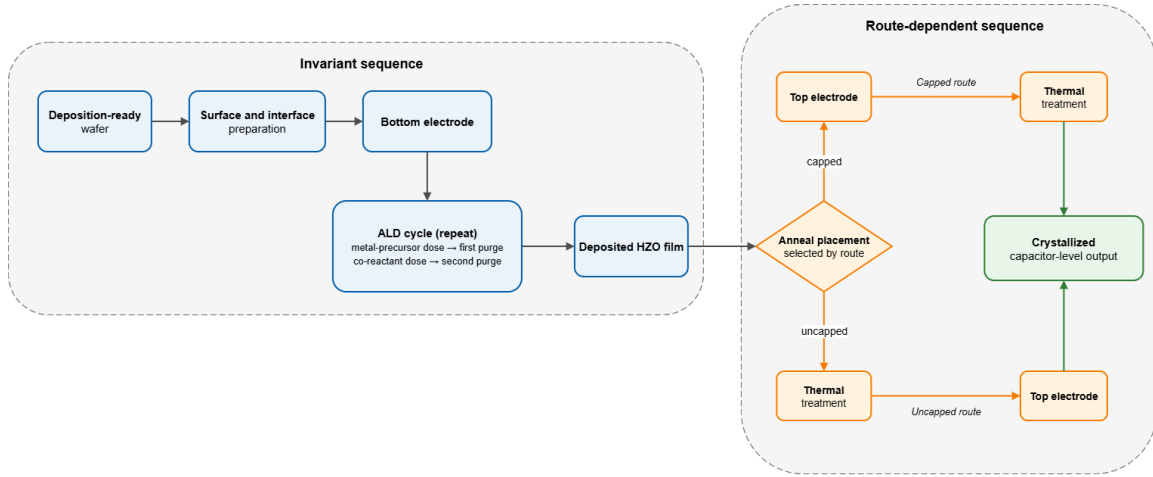
A TiN/HZO/TiN route records a 1:1 $\text{HfO}_2:\text{ZrO}_2$ cycle ratio for nominal $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ [9]. Its ratio sits inside the documented architectures and matches one W/HZO/W route approximately, yet the 3:2 PEALD example proves that nominal equimolar composition does not entail 1:1 cycling [3, 11].

Table 3.2: Documented ALD cycle chemistries and operating windows

Film/stack	Hf precursor	Co-reactant	Cycle/timing	T/P window	Source/condition
HfO_2	Cp_2HfMe_2 ; $(\text{MeCp})_2\text{HfMe}_2$; $(\text{MeCp})_2\text{Hf}(\text{OMe})\text{Me}$; $\text{Hf}(\text{NEtMe})_4$; $\text{Hf}(\text{NMe}_2)_4$	H_2O , O_2 plasma, O_3	Not covered in sources	250–300 °C stability	Precursor and oxidant survey [5]
$\text{HfO}_2/\text{ZrO}_2$	TEMAHf	H_2O , O_3 , O_2 plasma, H_2O_2	DF-ALD repeats precursor half-cycle	Not covered in sources	DF-ALD review [6]
TiN/10 nm $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ / TiN	Not covered in sources	H_2O	56–68 supercycles	120–250 °C	Thermal ALD chamber study [10]
10 nm $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$	TEMA-Hf	O_2 plasma, 3–6 s	1.0/1.5 s doses; 3–90 s Ar purge	260 °C; 80 mTorr	3:2, 17-supercycle PEALD study [11]
$\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$	TDMA-Hf	O_3 , 400 g m^{-3}	Not covered in sources	Not covered in sources	High-concentration O_3 ALD [1]
$\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ nanolaminate	TEMA-Hf	O_2 plasma, 200 W	$(5, 5) \times 12 - (60, 60) \times 1$; 120 cycles	250 °C	Seed-layer PEALD study [7]
$\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$	Not covered in sources	H_2O	N_2 second purge	200 °C	Thermal ALD study [8]
10 nm HfO_2 / SiO_2	HfCl_4	H_2O	10:1–27:1 $\text{HfO}_2:\text{SiO}_2$ cycles	300 °C	Thermal HSO process [13]
Hybrid HfO_2 / SiO_2	Not covered in sources	O_2 , Ar plasma	1 SiO_2 :16–32 HfO_2 cycles	50 °C SiO_2 PEALD	Hybrid HSO process [13]
5.5 nm SS-/SL- $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$	TDMAHf	H_2O	SS $(1 + 1) \times 27$; SL $5(5 + 6) + 5$	200 °C	Thermal superlattice ALD [2]
$\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ / ZrO_2 / $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$	TEMAH	Oxygen plasma	Not covered in sources	250 °C	Zr-rich stack PEALD [4]
10 nm $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ on W	Not covered in sources	O_3 , 276 g Nm^{-3}	1:1 Hf:Zr; 2–15 s O_3 pulse	250 °C; 90 °C precursors	O_3 -pulse ALD study [3]

3.7 Post-Deposition Thermal Treatment

Thermal treatment converts the deposited film toward the route’s crystalline state and can also terminate the usable process window. A 650 °C seed-layer anneal increases monoclinic phase and degrades ferroelectric behavior, while W/HZO/W annealing above 720 °C causes large leakage and short circuits [3, 7]. Temperature, time, ambient, and electrode boundary must therefore be evaluated as one treatment event.

Figure 3.1: ALD HfO₂ module process flow

Electrode state determines where the treatment is placed. PDA is recorded without a TiN cap or before the upper electrode, PMA follows capping or the top electrode, and a deliberate capped–uncapped comparison applies RTP on opposite sides of top-electrode deposition [5, 7, 8, 10]. A wafer-scale route anneals after top-TiN patterning but before bottom-TiN localization, while another pre-anneals the bottom TiN before HZO specifically to isolate top-electrode stress [1, 2]. The sub-400 °C BEOL-compatible route is a module limitation; higher-temperature examples do not override its stated budget [6, 13].

The TiN/HZO/TiN route records RTP at 500 °C for 30 s in N₂ [9]. The complete temperature–time–ambient tuple exactly matches a TiN/HZO/TiN post-metallization condition reported by Choi et al. [10]. This is the strongest cross-route corroboration because stack, temperature, duration, and ambient agree without averaging. Thirteen sources cover thermal treatment, but only one reports a heating ramp; ramp profile remains the section-level evidence gap.

Table 3.3: Reported post-deposition treatment conditions

Film/stack	Anneal sequence	Temperature	Time	Ambient	Source/condition
HfO ₂	PDA; uncapped RTA	Not covered in sources	Not covered in sources	Not covered in sources	PDA definition [5]
HfO ₂	PMA after TiN capping	Not covered in sources	Not covered in sources	Not covered in sources	PMA definition [5]
HfO ₂ –ZrO ₂	Low-temperature crystallization	< 400 °C	Not covered in sources	Not covered in sources	BEOL compatibility [6]
Hf _{0.5} Zr _{0.5} O ₂	PMA by RTA	400–700 °C	30 s	N ₂	Temperature series [10]
Hf _{0.5} Zr _{0.5} O ₂	RTA; 50 °C/s ramp	600 °C	30 s	N ₂ , 1 atm	PEALD capacitor stack [11]
TiN/ Hf _{0.5} Zr _{0.5} O ₂ / TiN	PMA by RTA	400 °C	60 s	N ₂	Top-electrode anneal [1]
TiN/ Hf _{0.5} Zr _{0.5} O ₂ / TiN	Forming-gas furnace anneal	400 °C	30 min	5% H ₂ + 95% N ₂	Hydrogen treatment [1]
Hf _{0.5} Zr _{0.5} O ₂	PDA before upper electrode	450–650 °C	30 s	N ₂	Seed-layer study [7]
Ni-capped Hf _{0.5} Zr _{0.5} O ₂ MIM	RTP after top-electrode deposition	823 K	30 s	N ₂	Capped structure [8]

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Table 3.3: Reported post-deposition treatment conditions (*continued*)

Film/stack	Anneal sequence	Temperature	Time	Ambient	Source/condition
Uncapped Hf _{0.5} Zr _{0.5} O ₂ MIM	RTP before top-electrode deposition	Not covered in sources	Not covered in sources	Not covered in sources	Uncapped comparison [8]
Si:HfO ₂	Anneal after stack formation	800 °C	20 s	N ₂	Polar-phase formation [13]
TiN/HfO ₂ /ZrO ₂ /TiN	After TE patterning	470 °C	30 s	N ₂	Wafer-scale capacitors [2]
Conventional and Zr-rich Hf _{0.5} Zr _{0.5} O ₂	Rapid thermal annealing	400–450 °C	60 s	N ₂	Stack comparison [4]
W/ Hf _{0.5} Zr _{0.5} O ₂ / W	Rapid annealing	500–800 °C	1–30 s	N ₂	15 s O ₃ dosage [3]
HfO ₂ capacitor	High-pressure oxygen anneal	Not covered in sources	Not covered in sources	High-pressure O ₂	Wake-up suppression [12]
HfO ₂ capacitor	NH ₃ plasma pre-treatment	Not covered in sources	Not covered in sources	NH ₃ plasma	Endurance strategy [12]
Hf _{0.5} Zr _{0.5} O ₂ /TiN	Sequential O ₂ –H ₂ plasma oxidation	Not covered in sources	Not covered in sources	O ₂ –H ₂ plasma	TiO _x N _y growth [6]

3.8 Module Output

The module output is a capacitor-level material state, not a free-standing film measurement. It contains the HfO₂-based layer at its reported composition and thickness, the lower and upper electrode boundaries present at handoff, and the completed thermal treatment where the route places it. The output record must retain the ALD chemistry, purge and exposure history, electrode positions, anneal tuple, and any deliberately introduced interface or seed layer because those inputs direct oxygen vacancies, stress, phase formation, and endurance [1, 7, 10, 11].

The target phase state is an orthorhombic ferroelectric contribution in a working stack. GI-XRD reports the orthorhombic (111) reflection near $2\theta = 30.4^\circ$, STEM directly confirms orthorhombic material, and HRTEM shows large orthorhombic grains in one W/Pt/HZO/W route [3, 5]. That output definition does not require a claim of phase purity: metastable diffraction peaks can overlap, instrumental resolution can hide monoclinic material, and ALD hafnia can remain polycrystalline with nonferroelectric phases [3, 5, 11].

Functional handoff additionally requires evidence that the stack has not crossed a leakage or breakdown boundary. Excess crystallization creates a leakage path in one seed-layer series, and W/HZO/W devices short above the reported 720 °C limit [3, 7]. Polarization and endurance values remain qualification attributes of the same route rather than universal output thresholds. This section synthesizes output evidence distributed across phase, interface, electrode, thermal-treatment, leakage, polarization, and reliability targets; it does not pool those targets into a single recommended specification. The evidence does not define an array-level handoff state, so the output boundary stops at the capacitor stack.

3.9 Materials, Reactor Platforms, and Equipment

Reactor geometry sets the transport context for every dose and purge. The evidence includes thermal and plasma-enhanced ALD, single-wafer processing, a crossflow reactor, a showerhead reactor, and a laminar-flow side-inlet configuration; their thickness patterns and operating conditions remain attached to those platforms [2, 10, 13]. Named deposition tools include Oxford FlexAL II, iOV dx2, and TFS 200 Beneq for specified PEALD cases [4, 7, 11].

Specimen format determines what the equipment result represents. A $6400\ \mu\text{m}^2$ patterned electrode is a capacitor specimen, while a 150 mm wafer with 25–100 μm capacitors supports a spatial uniformity study; neither descriptor substitutes for the other [2, 10]. Thickness and composition equipment likewise form a measurement chain: ellipsometry controls reference-film thickness, XPS measures unannealed-surface stoichiometry, and paired Radiant/Agilent instruments measure the electrical stack [4, 8]. Eight sources cover the materials-equipment target, but several thermal, single-wafer, and laminar-flow reports omit manufacturer or model, and one named Savannah S100 has no recorded operating condition [1, 2, 8].

Table 3.4: Materials, reactor platforms, and measurement equipment

Equipment class	Tool/reactor	Application	Wafer/film format	Source/condition
Deposition reactor	Thermal ALD reactor	HZO deposition	$6400\ \mu\text{m}^2$ electrodes	HZO deposition process; patterning [10]
Deposition reactor	Oxford FlexAL II	PE-ALD HZO deposition	Not covered in sources	PE-ALD HZO film [11]
Deposition reactor	Savannah S100 ALD	Not covered in sources	Not covered in sources	Condition unspecified [1]
Deposition reactor	iOV dx2 PEALD	HZO thin-film growth	50 nm TiN on oxide/silicon	iSAC growth; TiN sputtering [7]
Deposition reactor	Single-wafer ALD reactor	HZO thin-film growth	Single-wafer processing	HZO growth methods [8]
Deposition reactor	Laminar-flow side-inlet ALD reactor	SS- and SL-HZO growth	150 mm wafers	SS/SL comparison [2]
Deposition reactor	TFS 200 Beneq PEALD	Stack growth	HZO/ZrO ₂ /HZO stack	250 C substrate [4]
Annealing equipment	Allwin21 AccuThermo AW 610	Rapid thermal annealing	HZO film	Rapid thermal anneal [11]
X-ray metrology	Rigaku SmartLab X-ray diffractometer	GIXRD and XRR	HZO film	Cu K-alpha radiation [11]
Thickness/composition metrology	Spectroscopic ellipsometry; XPS	Thickness control; stoichiometry monitoring	ALD HZO; unannealed samples	Methods measurements [8]
Electrode deposition	PVD 75 sputtering system	Bottom and top W deposition	Not covered in sources	W-electrode sputtering [4]
Electron microscopy	Helios G4 FIB; Talos F200x	Lamella preparation; HR-TEM/STEM	Not covered in sources	30–200 kV operation [4]
Electrical characterization	Radiant Workstation; Agilent B1500	Ferroelectric and parameter measurements	Not covered in sources	Paired electrical instruments [4]

3.10 Critical Process and Device Parameters

3.10.1 Film Thickness and Growth per Cycle

Growth per supercycle decreases as chamber temperature rises in the 10 nm thermal-HZO series. The route therefore increases its supercycle count from 56 to 68 between 120 and 250 °C rather than treating count as a fixed thickness control [10]. Independent HfO₂ and ZrO₂ calibrations use different TEMA- or TEMAH/TEMAZ-based chemistries, so their reported growth-per-cycle values remain source-specific calibrations [3, 7, 11].

Thickness provenance determines whether a value describes the device or a companion specimen. Ellipsometry on Si reference pieces supplies the SS/SL thickness, whereas cross-sectional imaging measures an actual capacitor stack in other routes [2, 10]. For Si-containing

hafnia, thermal and plasma-enhanced SiO₂ doping layers also have different deposition rates, so the nominal cycling ratio does not erase the process-mode effect [13].

3.10.2 Conformality and Uniformity

Conformality and wafer uniformity are different outcomes. ALD supplies comparative step coverage for a 10 nm ferroelectric-HfO₂ capacitor in a 13:1 three-dimensional geometry, but that result does not measure radial wafer variation [5]. In Si:HfO₂, similar thickness standard deviations conceal different spatial patterns: the thermal crossflow route decreases from inlet to outlet, while the hybrid showerhead route forms an edge-enhanced ring [13].

Wafer-scale electrical uniformity depends on the sampling and failure attribution. The SS/SL study separates a 70×70 mm² center region from the full 150 mm wafer, samples 140 capacitor sites, and attributes one failed SL point to incomplete Ti/Au removal rather than the HZO film [2]. Thirteen claims from only three sources cover conformality and uniformity; eleven sources provide neither a spatial sampling plan nor a wafer-uniformity result.

3.10.3 Crystalline Phase and Microstructure

Electrode boundary and processing history determine the reported phase mixture. TiN deposited at 500 °C produces confirmed monoclinic material, whereas room-temperature TiN suppresses monoclinic formation through stress-induced crystallization; annealing HZO before the top electrode gives an orthorhombic (111) fraction of about 42.9% on TiN and 29.7% on SiO₂ [1]. A Ru-bottom-electrode specimen separately produces a stronger orthorhombic peak than its Si-substrate comparison [8]. These phase fractions and peak trends belong to their electrode configurations.

The orthorhombic assignment has a reported diffraction signature and a reported ambiguity. The (111) reflection appears near $2\theta = 30.4^\circ$, but similar d -spacings, fine-grain broadening, and uncertain biaxial strain prevent three metastable phases from being separated in one GIXRD series [5, 11]. No monoclinic peak is detected after RTA in the W-capped route, yet the source limits that absence to the instrument’s resolution [3]. A nondetection therefore remains a bounded result rather than a phase-purity assertion.

Precursor-purge duration shifts microstructure within one PEALD series. Processed films carry 3.0–3.9 GPa tensile stress, while mean grain size decreases from 18.2 ± 0.1 to 16.2 ± 0.1 nm between the 3 and 90 s purge specimens [11]. The stress span is a within-series property window, and the grain-size direction is not generalized to other reactors.

Bulk phase properties provide context rather than an ALD operating window. At room temperature, bulk tetragonal and cubic HfO₂ are reported with permittivities of 46 and 36, and their monoclinic-to-tetragonal and tetragonal-to-cubic transitions occur at 1973 and 2773 K at 1 atm [14]. Thin-film ferroelectricity is separately reported for Si-, Y-, Al-, and Zr-doped hafnia and for properly processed undoped films, with reproducibility over 5–30 nm [14]. Those bulk transition temperatures are material properties, not crystallization recommendations for the capacitor module.

Phase-linked functional properties remain specimen-specific. The Pca2₁ phase has a theoretical remanent-polarization value of about 51 $\mu\text{C}/\text{cm}^2$, a 20 nm Si:HfO₂ film has a reported positive d_{33} of 17.8 pm/V, and a calculated ferroelectric-HfO₂ band structure has a Dresselhaus parameter of 0.578 eV Å [6, 12]. In the W-capped route, raising ozone pulse duration from 2 to 15 s increases $2P_r$ from 37 to 44 $\mu\text{C}/\text{cm}^2$, and HRTEM of the W/Pt/HZO/W instance shows large orthorhombic grains [3]. These outcomes are properties of the stated phase and stack, while their measurement techniques remain in Section 3.11.2.

3.10.4 Polarization and Coercive Field

Deposition temperature directs polarization through the vacancy state. In TiN/HZO/TiN, higher chamber temperature increases oxygen-vacancy indicators; those vacancies raise remanent polarization but lower the critical fatigue-onset cycle, while the 120 °C film has significantly lower P_r than the 200 and 250 °C films [10]. Anneal-temperature comparisons in that series therefore require a matched chamber temperature.

Seed position separates polarization states at fixed nominal nanolaminate architecture. At 550 °C, unseeded and top-seeded (20, 20) × 3 specimens form one response pair, while bottom- and dual-seeded specimens form another; at 650 °C, monoclinic formation degrades the response [7]. Increasing nanolaminate thickness raises remanent polarization but reduces durability in the same study [7].

Conditioning and readout define which polarization value is being asserted. W/HZO/W at a 15 s ozone pulse and 700 °C/5 s has distinct pristine, woken, and PUND-corrected values, and inserting Pt changes the PUND-corrected result under the same anneal [3]. Sub-6-nm HZO/Zr-RL/HZO reports its $2P_r$ /coercive-voltage pair at 1.0 V and retains a stable temperature response from 25 to 125 °C for that stack [4]. These states are not interchangeable measurement labels.

3.10.5 Leakage-Current Constraints

Leakage rises with both anneal temperature and chamber temperature in the matched TiN/HZO/TiN series. The direction is measured at −3 V by changing one temperature while holding the other fixed, which makes the two process couplings explicit [10]. This trend does not establish an absolute leakage limit outside that stack.

Bias and conditioning distinguish the other leakage results. The SS/SL wafer comparison reports post-wake-up maxima at ±3.0 V, whereas the 12 V current on a $1.1 \times 10^{-3} \text{ cm}^2$ electrode is identified as domain-switching current rather than leakage [2, 8]. Eight leakage-parameter claims come from four sources; ten sources provide no target-level leakage value, and none supplies a cross-route matched-bias comparison.

3.10.6 Defect Density, Endurance, and Reliability

Precursor-purge duration moves the breakdown boundary by more than an order of magnitude in one HZO route. A 3 s purge survives beyond 10^9 cycles, while the 15, 30, 60, and 90 s specimens break at 2×10^8 , 2×10^8 , 10^8 , and 6×10^7 cycles, respectively [11]. The same series links reduced purge to retained ligands, so chemistry and survival remain coupled even though the impurity signal alone does not establish the breakdown mechanism.

Chamber temperature trades polarization against fatigue onset through oxygen vacancies. All tested chamber/anneal combinations survive to 10^8 cycles, but higher chamber temperature lowers the cycle count at which fatigue begins while raising P_r [10]. This is a degradation-onset change, not a contradiction with the common 10^8 -cycle breakdown observation.

Architecture and stress amplitude set separate reliability trajectories. At matched voltage, SL outlasts SS, increasing cycling voltage shortens both lives, and post-wake-up retention loss stays below 4% through 10^4 s after 10^6 cycles [2]. In sub-6-nm HZO/Zr-RL/HZO, endurance approaches 10^{11} cycles at no more than 1.5 V, the 1.0 V result is breakdown-limited, and the 10-year retention projection belongs to 1.0 V at 125 °C [4]. No common stress waveform spans these route families, which is the section-level evidence gap.

Table 3.5: Critical process and device parameter windows

Parameter	Range/result	Film/stack	Process condition	Limitation	Source
HfO ₂ growth rate	1.00–1.18 Å/cycle	Independent HfO ₂ films	TEMA-Hf or TEMAH precursor processes	Not covered in sources	[3, 7, 11]
ZrO ₂ growth rate	1.00–1.34 Å/cycle	Independent ZrO ₂ films	TEMA-Zr or TEMAZ precursor processes	Not covered in sources	[3, 7, 11]
Film thickness	5.5–25 nm	Hf _{0.5} Zr _{0.5} O ₂ and SS/SL HZO	ALD; ellipsometry reference pieces	Not covered in sources	[2, 8]
Si-doping-layer deposition rate	PEALD significantly exceeds thermal ALD	Hybrid versus standard HSO	PEALD versus thermal ALD	Not covered in sources	[13]
Conformal high-aspect coverage	13:1 aspect ratio	10 nm FE-HfO ₂ capacitor	3D 1T-1C FeRAM	Not covered in sources	[5]
Step coverage	Excellent for high-aspect structures	ALD HfO ₂ films	ALD versus PVD/CVD	Polycrystalline films include nonferroelectric phases	[5]
Process-dependent thickness	10.6–12.7 nm	Standard HSO; hybrid HSO	All tested cycling ratios	Not covered in sources	[13]
Thickness nonuniformity	3.2–3.7% standard deviation	Standard and hybrid HSO	Thermal ALD and hybrid PEALD	Not covered in sources	[13]
Effective sublayer thickness	0.5 nm	SL-HZO: ZrO ₂ /HfO ₂	5 ZrO ₂ plus 6 HfO ₂ cycles	Not covered in sources	[2]
Wafer-scale uniformity improvement	18–53%	SS/SL HZO, 150 mm wafers	70×70 mm ² center; ±3.0 V	One SL site failed during lift-off	[2]
Maximum leakage-current density	0.04–0.08 A/cm ²	SS/SL HZO capacitors	±3.0 V after wake-up	Not covered in sources	[2]
Leakage-current trend	Increases with chamber and anneal temperature	TiN/HZO/TiN capacitor	-3 V; chamber 120–250 °C; anneal 400–700 °C	Not covered in sources	[10]
Domain-switching current	Up to 80 mA	Hf _{0.5} Zr _{0.5} O ₂ capacitor	12 V; 1.1×10 ⁻³ cm ² electrode	Not covered in sources	[8]
Pristine P _r	0.28–31.8 μC/cm ²	TiN/HZO/TiN capacitors	120–250 °C chamber; 400–700 °C anneal	120 °C significantly degrades P _r	[10]
E _c and 2P _r	±1.5 MV/cm; about 40 μC/cm ²	TiN/HZO/TiN 1T-1C capacitor	State-of-art FeRAM	Not covered in sources	[6]
E _c	0.8–2 MV/cm	Doped HfO ₂	Si, Al, Y, La dopants	Not covered in sources	[14]
Switched polarization charge	30–60 μC/cm ²	Doped HfO ₂	Si, Al, Y, La dopants	Not covered in sources	[14]
Residual polarization	23.94–28.18 μC/cm ²	HZO; none or dual ZrO ₂ seed	550 °C anneal	650 °C forms monoclinic phase	[7]
2P _r and coercive voltage	43.4 μC/cm ² ; 0.75 V	Sub-6 nm HZO/Zr-RL/HZO	1.0 V operation	Not covered in sources	[4]
Pristine and woken 2P _r	52–65 μC/cm ²	W/HZO/W; W/Pt/HZO/W	700 °C, 5 s; 15 s O ₃	Not covered in sources	[3]

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Table 3.5: Critical process and device parameter windows (*continued*)

Parameter	Range/result	Film/stack	Process condition	Limitation	Source
Purge-time endurance	6×10^7 to $> 10^9$ cycles	Hf _{0.5} Zr _{0.5} O ₂ capacitors	3–90 s precursor purge	Longer purge reduced endurance	[11]
Temperature-controlled fatigue	Earlier onset at higher chamber temperature	TiN/HZO/TiN capacitors	120–250 °C chamber	Vacancies raise P_r but reduce cycles	[10]
High-field breakdown cycles	SS 10^5 ; SL 4×10^5	SS/SL HZO capacitors	± 3.5 V cycling	Not covered in sources	[2]
Moderate-field endurance	SS about 5×10^6 ; SL $> 10^7$	SS/SL HZO capacitors	± 3.0 V cycling	SS fails earlier	[2]
Retention loss	Below 4% through 10^4 s	SS/SL HZO capacitors	After 10^6 cycles	SL loss slightly lower	[2]
Endurance and retention	About 10^{11} cycles; 82% after 10 years	Sub-6 nm HZO/Zr-RL/HZO	1.0–1.5 V; 125 °C	Endurance is breakdown-limited	[4]

3.11 Electrical Qualification and Metrology

3.11.1 Film-Thickness and Uniformity Measurement

Ellipsometry and cross-sectional TEM test different parts of the thickness assertion. The corresponding process properties are analyzed in Sections 3.10.1 and 3.10.2. Ellipsometry measures film and seed-layer thickness over a defined sampling pattern, while TEM confirms the ferroelectric layer in an actual cross-section and tests invariance across chamber-temperature conditions [7, 10]. Multipoint ellipsometry samples 49 wafer positions with a 3 mm edge exclusion, revealing an inlet-to-outlet gradient in the thermal crossflow process and an edge ring in the hybrid showerhead process despite similar standard deviations [13].

Reference-piece measurements do not become device cross-sections. The SS/SL study derives thickness from Si pieces grown with the characterized wafers, so its value carries the companion-specimen relation into the result [2]. Sampling pattern, edge exclusion, reactor geometry, and specimen identity are therefore required measurement properties.

3.11.2 Crystalline-Phase and Structural Characterization

GI-XRD and STEM supply complementary phase evidence. The corresponding crystalline properties are analyzed in Section 3.10.3. GI-XRD assigns diffraction peaks over the illuminated film volume, whereas STEM directly images local orthorhombic material and can follow an interfacial monoclinic-to-orthorhombic transition during wake-up [5]. Similar d -spacings, fine-crystallite broadening, and uncertain biaxial strain make three metastable phases inseparable in one GIXRD series, while finite XRD resolution limits any inference from a missing monoclinic peak [3, 11].

Two-dimensional XRD and FTIR-ATR measure different structural correlates. The former derives processed-film stress through $\sin^2 \psi$ analysis; the latter records an absorption spectrum whose sharp 670 cm^{-1} feature is identified as an artifact [11]. STEM also tests continuity and thickness, but bottom-TiN roughness near the scale of a 5.5 nm superlattice prevents HfO₂/ZrO₂ sublayers from being resolved in that lamella [2]. Technique, geometry, resolution, and cross-sectional specimen condition remain part of each result.

3.11.3 Hysteresis, Cycling, and Reliability Testing

PUND separates switched polarization from leakage and dielectric contributions, while P–E or P–V loops record the hysteresis response under a stated waveform and frequency [1, 5, 6]. The corresponding outcome properties are analyzed in Sections 3.10.4 and 3.10.6. Designed pulse trains add switching-dynamics information rather than another loop value [4]. The measurand therefore determines which electrical protocol is admissible.

Cycling and readout order reveal wake-up and fatigue as a trajectory. A seed-layer study applies rectangular cycling before triangular-pulse measurement; its bottom- and dual-seeded devices wake up before fatigue after the polarization maximum, whereas unseeded and top-seeded devices fatigue without wake-up [7]. The wafer-scale route interleaves defined up/down pulses, breaks, and triangular PUND sweeps during or after wake-up [2].

Retention and ultrathin switching inherit their electrical setup. One retention test uses 12 V, 600 ns pulses on a $1.1 \times 10^{-3} \text{ cm}^2$ electrode with severe leakage, while verification below 5 nm is described as leakage dominated [8, 14]. Waveform, field, frequency, pulse width, area, conditioning state, and readout method remain part of the qualification result.

3.11.4 Leakage and Breakdown Measurement

Sweep failure, spatial leakage mapping, and time-zero dielectric breakdown answer different questions. The corresponding leakage property is analyzed in Section 3.10.5. The seed-layer route uses a 0–15 V DC sweep in 0.5 V steps, whereas the SS/SL wafer maps leakage at $\pm 3.0 \text{ V}$ against radial bottom-TiN thickness [2, 7]. The former connects excessive crystallization to a leakage path; the latter connects electrical spatial variation to an electrode-thickness coordinate.

Thermal failure requires the stack and anneal tuple. W/HZO/W develops large leakage and shorts above 720°C , while the reported TZDB comparison changes stack and anneal but omits waveform, electrode area, and stress time [3, 4]. Four sources contribute fourteen leakage-breakdown claims; the other ten provide no target-level leakage or breakdown method.

3.11.5 Interface and Defect Characterization

Chemical depth profiles and cross-sectional images form a joint interface description. The corresponding defect and reliability property is analyzed in Section 3.10.6. O 1s XPS separates lattice-oxygen and sub-oxide components, EDS supplies Hf/Zr/O fractions, D-SIMS profiles oxygen and bottom-TiN oxidation, and ToF-SIMS tracks carbon and nitrogen retained after shortened purges [1, 10, 11]. HRTEM then distinguishes the W/HZO interfacial region from a sharp Pt/HZO interface, while HR-STEM provides atomic-scale lamella imaging [3, 4].

Depth sensitivity limits the electrode state that can be characterized. Laboratory XPS reaches about 5 nm and precludes a realistic top electrode, whereas HAXPES gains one to two orders of magnitude in depth for nondestructive profiling through electrodes [6]. Adventitious carbon and absorbed OH confound XPS-derived oxygen stoichiometry, and ToF-SIMS does not quantify hydrogen ions [11]. Morphology and chemical depth information remain complementary evidence objects.

3.11.6 Electrical Stressing and Qualification Protocols

Conditioning state determines the electrical state presented to readout. One protocol applies 5×10^3 square-wave wake-up cycles around its measurements, while another applies 10^3 cycles before PUND at different amplitude and frequency [4, 11]. Those counts define different conditioned individuals in the process record.

Waveform and electrode reference determine the applied stress. Reported endurance uses either 3 V trapezoids with 1 μs ramps and holds or 2.0 MV/cm bipolar pulses at 10 kHz with 2 μs width and 20 ns edges [10, 11]. One setup biases the TiN top electrode with the bottom

grounded, while another biases the bottom electrode with the top grounded [1, 4]. Conditioning state, waveform shape, amplitude or field, frequency, timing, cycle count, and driven electrode are therefore explicit qualification properties.

Table 3.6: Qualification and metrology method-to-output matrix

Test/technique	Measurand/output	Specimen/stack	Stress/setup	Limitation	Source
GI-XRD	Orthorhombic peak at 30.4°	HZO thin film	Grazing-incidence diffraction	Metastable phase peaks overlap	[5, 11]
2D XRD	Tensile stress 3.0–3.9 GPa	Processed HZO films	18° incidence	Not covered in sources	[11]
FTIR-ATR	Absorption from 400–900 cm ⁻¹	HZO purge-series films	4 cm ⁻¹ resolution	670 cm ⁻¹ peak is artifact	[11]
STEM	Orthorhombic phase and wake-up transition	Ferroelectric hafnia film	Interfacial cycling observation	Not covered in sources	[5]
Theta–theta XRD	Ru electrode strengthens orthorhombic peak	Ni/HZO/Ru and Ni/HZO/Si	Bottom-electrode comparison	Not covered in sources	[8]
Spectroscopic ellipsometry	Thickness 12.64–13.37 nm	Unseeded HZO nanolaminates	Elli-SE-U measurement	Thickness varies with nanolaminate period	[7]
Multipoint spectral ellipsometry	Mean 10.76–12.74 nm; STD 3.19–3.26%	Standard and hybrid HSO	49 points; 3 mm exclusion	Patterns depend on reactor geometry	[13]
Cross-sectional TEM	Thickness invariant across chamber temperatures	TiN/HZO/TiN capacitors	120–250 °C; 500 °C anneal	Not covered in sources	[10]
O 1s XPS	Binding peaks span 532–534 eV	HZO layer	700 °C anneal	Not covered in sources	[10]
XPS	Hf:Zr ratios 74:100–79:100	Bare HZO surfaces	3–90 s purge	Surface overlayers confound stoichiometry	[11]
ToF-SIMS	Carbon and nitrogen impurity profiles	PEALD HZO films	Dual-beam depth analysis	Hydrogen is not quantitative	[11]
D-SIMS	Oxygen profile; bottom-TiN oxidation	TiN/HZO/TiN stack	250 °C deposition; Cs source	Not covered in sources	[1]
HAXPES	Subsurface vacancy profile	Electroded oxide stack	Depth gain 1–2 orders	Classical XPS reaches about 5 nm	[6]
HRTEM	W-interface regions; sharp Pt interface	W/HZO/W and W/Pt/HZO/W	700 °C, 5 s anneal	Not covered in sources	[3]
Wake-up cycling	Conditioned ferroelectric response	TiN/HZO/TiN capacitors	10 ⁵ cycles; 2.5 MV/cm	Not covered in sources	[1]
Trapezoidal endurance cycling	Survival through 10 ⁸ cycles	TiN/HZO/TiN capacitors	3 V; 1 μs ramp/hold	Not covered in sources	[10]
Bipolar endurance waveform	Wake-up and endurance qualification	PEALD HZO capacitors	10 kHz; 2 μs; 2.0 MV/cm	Not covered in sources	[11]
Square-wave wake-up	Preconditioned PUND measurement	HZO; Zr-rich-layer stack	10 ³ cycles; 1.5 V; 1 kHz	Not covered in sources	[4]

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Table 3.6: Qualification and metrology method-to-output matrix (*continued*)

Test/technique	Measurand/output	Specimen/stack	Stress/setup	Limitation	Source
DC breakdown sweep	Breakdown 7.15–4.08 MV/cm	Dual-seeded HZO nanolaminates	0–15 V; 550–650 °C	Excess crystallization creates leakage paths	[7]
Radial leakage mapping	Leakage versus bottom-TiN thickness	SS- and SL-HZO wafers	± 3.0 V	Electrode thickness varies radially	[2]
TZDB	Time-zero breakdown behavior	HZO; Zr-rich-layer stack	Multiple annealing temperatures	Not covered in sources	[4]
Elevated-temperature I-V	Leakage reduced 20-fold	W/HZO/W and W/Pt/HZO/W	200 °C Schottky analysis	Above 720 °C causes shorts	[3]
PUND	Leakage-corrected switched polarization	Thin ferroelectric films	Positive-up negative-down pulse sequence	Depolarization fields cause underestimation	[5, 6]
P-E hysteresis	$2P_r$ and E_c	TiN/HZO/TiN capacitors	10 kHz	Not covered in sources	[1]
P-V hysteresis	Polarization before and after wake-up	HZO; Zr-rich-layer stack	1.75 V operation	Not covered in sources	[4]
Fatigue cycling	P_r falls to $5 \mu\text{C}/\text{cm}^2$	Capped 25 nm HZO	1 MHz; 8 V; 1000 ns	Inferior to SBT after 10^9 cycles	[8]
Seed endurance cycling	Stable through 10^7 – 10^8 cycles	Seeded HZO nanolaminates	550 °C; rectangular pulses	Fatigue follows 10^5 cycles	[7]
Wafer-scale PUND	$2P_r$ spans 27.6– $31.5 \mu\text{C}/\text{cm}^2$	SS- and SL-HZO capacitors	3.0 V; 150 mm wafer	RSD spans 2.2–7.0%	[2]
Wake-up cycling	Wake-up-free $2P_r$ near $64 \mu\text{C}/\text{cm}^2$	W/HZO/W capacitor	3 MV/cm; 100 Hz; 1000 cycles	Short ozone pulses pinch loops	[3]
Pulse-switching measurement	Switching dynamics	HZO/Zr-rich-layer/HZO	Designed pulse-train sequence	Not covered in sources	[4]

3.12 A Worked Route

The route in this section is one instantiation, not a recommendation or reference flow. Tsai et al. report an internally ordered TiN/HZO/TiN path through the literature windows without combining configurations [9].

Within this route, bottom-electrode formation spans both familiar and boundary-extending settings. The 15 nm TiN thickness and 100 W sputter power sit inside the documented electrode envelopes, while 2.5 mTorr extends the only prior sputter-pressure value downward; the Ar and N₂ flow rates introduce a quantity type not otherwise recorded. Their coexistence in one row of the route does not make them transferable to another sputter tool.

Within this route, the HZO deposition step exposes the most important unresolved condition. The 10 nm film, 250 °C deposition temperature, TEMA_H/TEMA_Z precursors, 90 °C TEMA_H source, and 1:1 cycle ratio sit inside the earlier windows, but the 110 °C TEMA_Z source lies beyond the previous 90 °C maximum [3, 9, 11]. Delivery mode and line temperatures are absent, so the extension cannot be reconciled. The oxidant is also unreported; the gap is retained even though the other cited routes name a co-reactant.

Within this route, pattern definition and wet etching belong to capacitor formation rather than ALD deposition proper. They appear here only to preserve sequence: photolithography is followed by an NH₄OH:H₂O₂:H₂O wet etch at the recorded 1:1:5 ratio and 60 °C, while detailed lithography parameters remain unreported [9]. These steps have no in-scope ALD window and

are excluded from the module taxonomy outside this worked example.

Within this route, the final 500 °C, 30 s, N₂ RTP tuple is inside the thermal window and exactly matches another reported TiN/HZO/TiN post-metallization condition [9, 10]. Agreement on stack, temperature, duration, and ambient is corroboration, not evidence that the tuple is preferred.

Table 3.7: One TiN/HZO/TiN route and its relation to the documented windows

Ordered step	Route state	Relation to earlier evidence	Scope or gap
1. Bottom electrode	15 nm TiN; sputter, 100 W, 50 sccm Ar, 3 sccm N ₂ , 2.5 mTorr	Thickness and power inside; pressure beyond prior low edge; gas-flow quantity new	In scope; tool identity unreported
2. HZO deposition	10 nm Hf _{0.5} Zr _{0.5} O ₂ at 250 °C; TEMA 90 °C; TEMA 110 °C; 1:1 cycles	Film, deposition temperature, precursors, Hf-source temperature, and ratio inside; Zr-source temperature beyond prior maximum	In scope; oxidant, pulse and purge timings, delivery mode, line temperatures, pressure, and count unreported
3. Top electrode	15 nm sputtered TiN	Thickness inside electrode envelope	In scope; position-specific sputter conditions unreported
4. Pattern definition	Photolithography	No ALD-module window	Outside deposition scope; detailed lithography parameters unreported
5. Wet etch	NH ₄ OH:H ₂ O ₂ :H ₂ O at 1:1:5 and 60 °C	No ALD-module window	Outside deposition scope; retained only for route order
6. Crystallization	RTP, 500 °C, 30 s, N ₂	Inside and exact independent match for the same stack	In scope; ramp rate unreported

Source: Tsai et al. [9].

3.13 Boundaries, Assumptions, and Limitations

3.13.1 Boundaries

The module boundary begins with the deposition-ready wafer region and ends with the capacitor-level film state defined in Section 3.8. In scope are surface and interface preparation, precursor selection and delivery, the two-dose/two-purge ALD cycle, cycle architecture, reactor and equipment context, deposition temperature, film composition and thickness, electrode boundary conditions, route-positioned thermal treatment, process controls, and the resulting film properties.

The chapter does not own the CMOS front end or downstream BEOL integration. Those operations are assigned to Chapters 2 and 4 respectively; they appear here only as the source and destination boundaries of the module, not as processes to be specified. A BEOL thermal budget can nevertheless constrain this module, as in the reported sub-400 °C route [6, 13].

Patterning is outside the chapter’s process taxonomy and is not owned elsewhere in this report. The photolithography and wet-etch operations appear only in the worked route so its order remains complete [9]. The chapter otherwise treats capacitor-level metal–ferroelectric–metal structures. Array-level and FeFET integration are excluded except when a source directly supplies a contextual claim, and such claims are not converted into capacitor-module requirements.

3.13.2 Assumptions

Ledger claims are recorded as reported. No value in this chapter was independently re-derived, re-measured, averaged, interpolated, or checked against a second reading of its source.

Single-source evidence is treated as evidence, not consensus. A setting reported by one study remains that study’s setting even when it falls inside a broader literature window.

No value is assumed transferable between configurations. Transfer requires the matching conditions defined in Section 3.3; missing conditions remain unknown under the open-world discipline.

The source set was assembled through a three-topic filter—HfO₂, FeRAM, and ALD—and is not an exhaustive literature survey. Functional definitions of the first and second purge are process-model assertions used to define the cycle operations; where the ledger does not report their duration, pressure, or failure mode, the chapter does not invent one.

3.13.3 Limitations

The evidence base is finite and uneven. It contains 754 claims from 14 sources, with 107 claims from the largest source and 17 from the smallest, and it contains no primary experimental data generated for this report. Claim count therefore measures documentation density rather than validity, independence, or process importance.

The weakest coverage is an evidence limitation. First purge has three claims from one source and second purge one claim from one source; input condition has seven claims from three sources; conformance/uniformity has thirteen claims from three sources; leakage-current parameters have eight claims from four sources; and leakage/breakdown metrology has fourteen claims from four sources. These thin targets do not imply that the corresponding operations or properties are weak in the physical module. They mean that the present source set documents them sparsely.

The worked route contains two gaps that block reconciliation. Its oxidant is unreported, and its 110 °C TEMAZ source temperature lacks delivery mode and line temperatures [9]. A PEALD route instead records an 85 °C TEMAZ bubbler with downstream temperatures of 120 and 140 °C, while another route records both precursor sources at 90 °C [3, 11]. Without the missing delivery metadata, the values remain parallel route instances rather than a resolved range.

Process constraints are module limitations, not evidence limitations. The sub-400 °C BEOL-compatible route limits which crystallization treatments can be used, and W/HZO/W shorts above 720 °C under its stated stack and anneal conditions [3, 6, 13]. Conversely, sparse second-purge coverage says nothing about whether a second purge is physically optional. This distinction is retained so later ontology metadata can separate process capability boundaries from open evidence gaps.

3.14 Ontology Model and Communication

3.14.1 Protégé Ontology Design

3.14.2 Classes and Properties

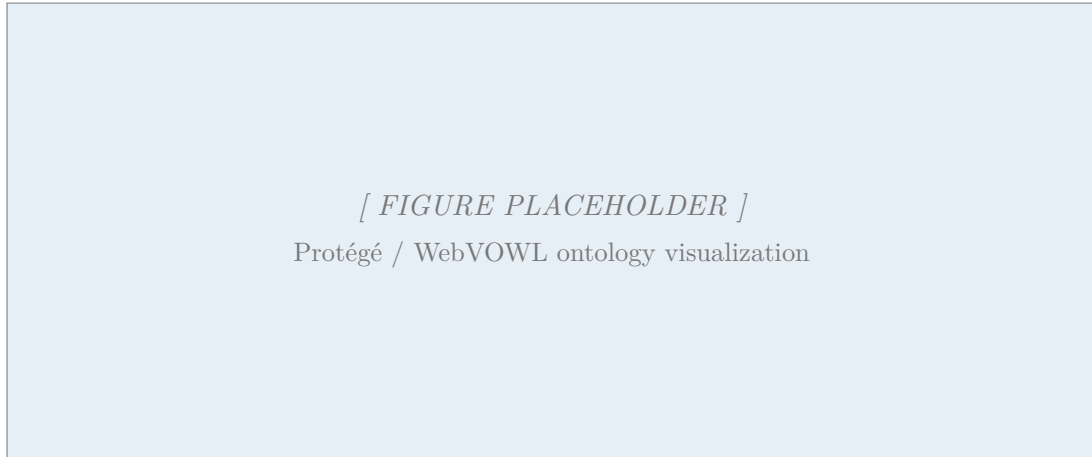
Table 3.8: ALD HfO₂ ontology classes and properties

Ontology element	Element type	Domain /parent	Range /definition

3.14.3 Individuals and Example Records

Table 3.9: ALD HfO₂ ontology individuals

Individual	Class	Key assertions	Source record
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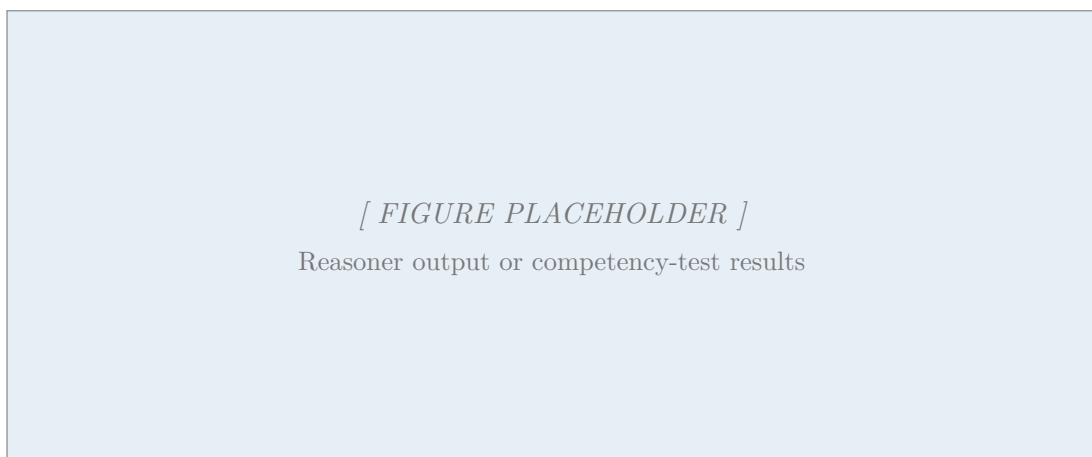
Figure 3.2: ALD HfO₂ ontology visualization

3.14.4 Digital Reference Mapping

Table 3.10: ALD HfO₂ Digital Reference mappings

Local concept	Digital Reference IRI	Mapping relation	Evidence and rationale
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3.14.5 Reasoner and Validation Results

Figure 3.3: ALD HfO₂ ontology validation results

3.14.6 Ontology Limitations

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